

Exercise 1C: Equilibrium and State Trajectory of an LTI System

Tag List: [STANDARD ASSESSMENT] [PARTIAL CREDIT] [SYMBOLIC] [CONTROL TOOLBOX]

Introduction

Given the LTI system described by means of the following state equations

$$\begin{cases} \dot{x}(t) = \begin{bmatrix} 2 & -5 & 0 \\ -1 & -2 & 0 \\ 0 & 0 & \alpha \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u(t) \\ y(t) = [1 \ 0 \ 1] x(t) \end{cases} \quad (*)$$

you are asked to

- assume $\alpha = -1$;
- determine the **equilibrium state** $\bar{x}$, corresponding to the constant input $\bar{u} = +1$, $\forall t \geq 0$;
- calculate the state trajectory $x(t)$ for the given system, when the applied input is $\bar{u}$, and the initial state is $x_0 = \bar{x} + \delta_x$, where δ_x is a suitable perturbation (more details into the assignment).

Assignment

Your code shall

Task 1

- define the symbolic parameter **p_alpha** and assign to it the symbolic value -1 (pretest);
- assign the symbolic matrix **symA** with the just assigned symbolic parameter **p_alpha** (pretest);
- convert the symbolic matrix **symA** into the numeric matrix **matA**; moreover, assign the numeric matrices **matB**, **matC** and **matD** according to the state equations described in [Eq, \(*\)](#);

Task 2

- assign the constant input **barU**, as a numeric variable;
- determine the **equilibrium state** $\bar{x}$ and assign the solution to the numeric variable **barX**;

Task 3

- use the preloaded variable **deltaX** to create a new initial state x_0 as

```
x0 = barX+deltaX;
```

- determine the **state trajectory** of the system, using **x0** as initial state and applying the constant input **barU**;
- compute the state trajectory for each time instant in the preloaded array **timeT** and assign the result to the numeric variable **varX**.

Solution

```
clear
load perturbX_C.mat
load timeT_C.mat

whos deltaX timeT
```

Name	Size	Bytes	Class	Attributes
deltaX	3x1	24	double	
timeT	1x20001	160008	double	

Task 1

```
p_alpha = sym(-1);

symA = [2, -5, 0;
```

```
-1, -2, 0;
0, 0, p_alpha]
```

```
symA =

$$\begin{pmatrix} 2 & -5 & 0 \\ -1 & -2 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

```

```
matA = double(symA)
```

```
matA = 3x3
     2     -5      0
    -1     -2      0
     0      0     -1
```

```
matB = [ 0;
        1;
        1]
```

```
matB = 3x1
     0
     1
     1
```

```
matC = [1, 0, 1]
```

```
matC = 1x3
     1      0      1
```

```
matD = 0;
```

Task 2

```
barU = +1;
```

```
% linear Eq. to solve:
% A*barX +B*barU = 0
% barX is the unknown variable
bVect = -matB*barU;
```

```
barX = linsolve(matA, bVect)
```

```
barX = 3x1
     0.5556
     0.2222
     1.0000
```

Task 3

```
x0 = barX+deltaX;
sysLTI = ss(matA, matB, matC, matD)
```

```
sysLTI =
```

```
A =
      x1  x2  x3
x1      2  -5   0
x2     -1  -2   0
x3      0   0  -1
```

```
B =
      u1
x1      0
x2      1
x3      1
```

```
C =
      x1  x2  x3
y1      1   0   1
```

```
D =
      u1
y1      0
```

Continuous-time state-space model.
Model Properties

```
[yT, ~, varX] = lsim(sysLTI, barU*ones(size(timeT)), timeT, x0);
```

Not requested:

```

figure;
plot(timeT, varX, 'LineWidth', 2);
zoom on; grid on;
hold on
plot(timeT, barX.*ones(size(barX,1),size(timeT,2)), 'LineStyle', '-.', 'LineWidth', 1.5)
xlabel('time [s]', 'fontsize', 14);
ylabel('state traj. x(t)', 'fontsize', 14);
legend('$x_1(t)$', '$x_2(t)$', '$x_3(t)$', '$\bar{x}_1$', '$\bar{x}_2$', '$\bar{x}_3$',...
'Location', 'bestoutside', 'Interpreter', 'latex', 'fontsize', 14);

```

